



Student Performance & Attendance Analysis

Cleaning, merging, and visualizing five linked classroom datasets

Problem Statement

?

Schools collect attendance, homework, and exam data in separate systems. That makes it hard to see, at a glance, how attendance and homework habits relate to exam performance — or to spot patterns by grade level and subject.

✓

This project cleans and merges five related datasets into one analysis-ready table, then visualizes subject-level performance, attendance patterns, and the relationship between attendance/homework and exam scores.

5

linked datasets

12,157

students covered

6

subjects tracked

The Dataset

Student Performance and Attendance Dataset — Marvy Ayman Halim, via Kaggle

<code>students.csv</code>	12,157	Roster: ID, name, grade level, date of birth
---------------------------	---------------	--

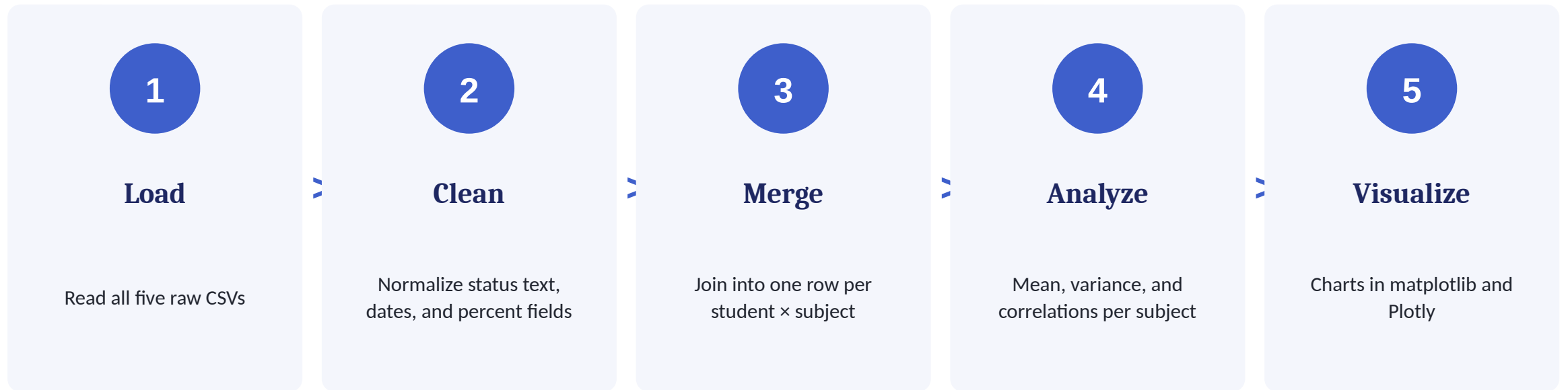
<code>attendance.csv</code>	~365,000	Daily attendance status by subject
-----------------------------	-----------------	------------------------------------

<code>homework.csv</code>	~61,000	Homework assignment status
---------------------------	----------------	----------------------------

<code>performance.csv</code>	~36,000	Exam scores & homework completion %
------------------------------	----------------	-------------------------------------

<code>teacher_parent_communication.csv</code>	~24,000	Communication log entries
---	----------------	---------------------------

Methods



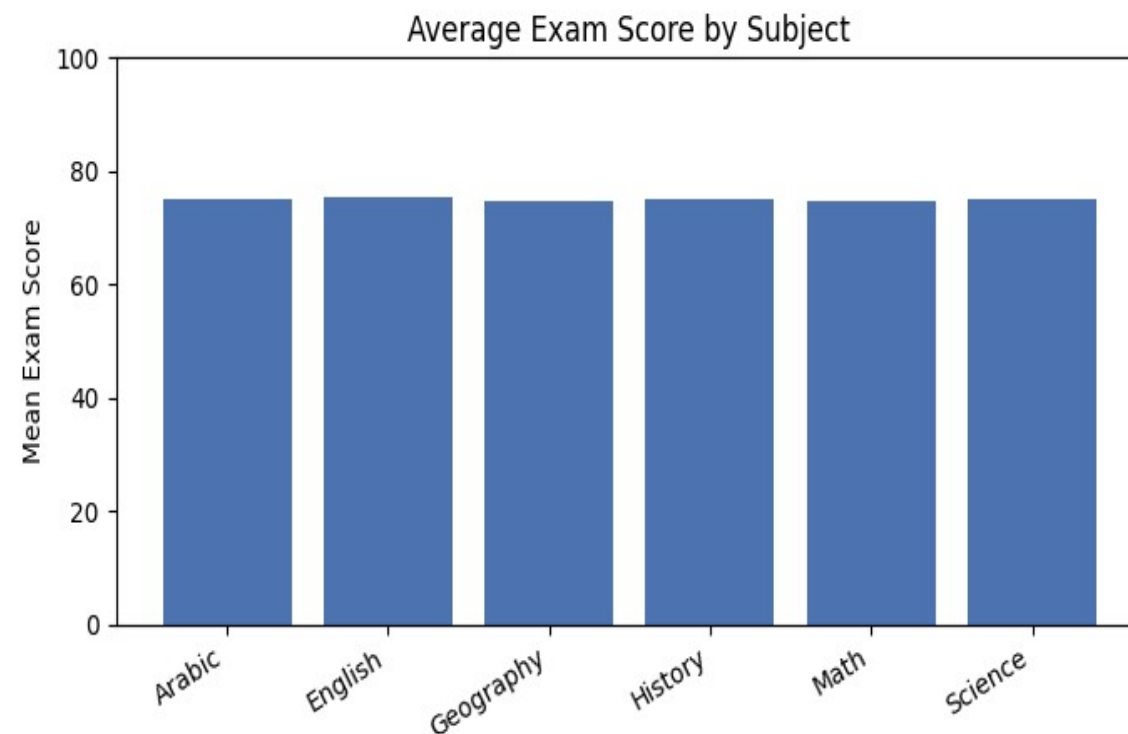
Full cleaning decisions documented in DATA_CLEANING_BRIEF.md and notebook section 3–4.

Results: Score by Subject

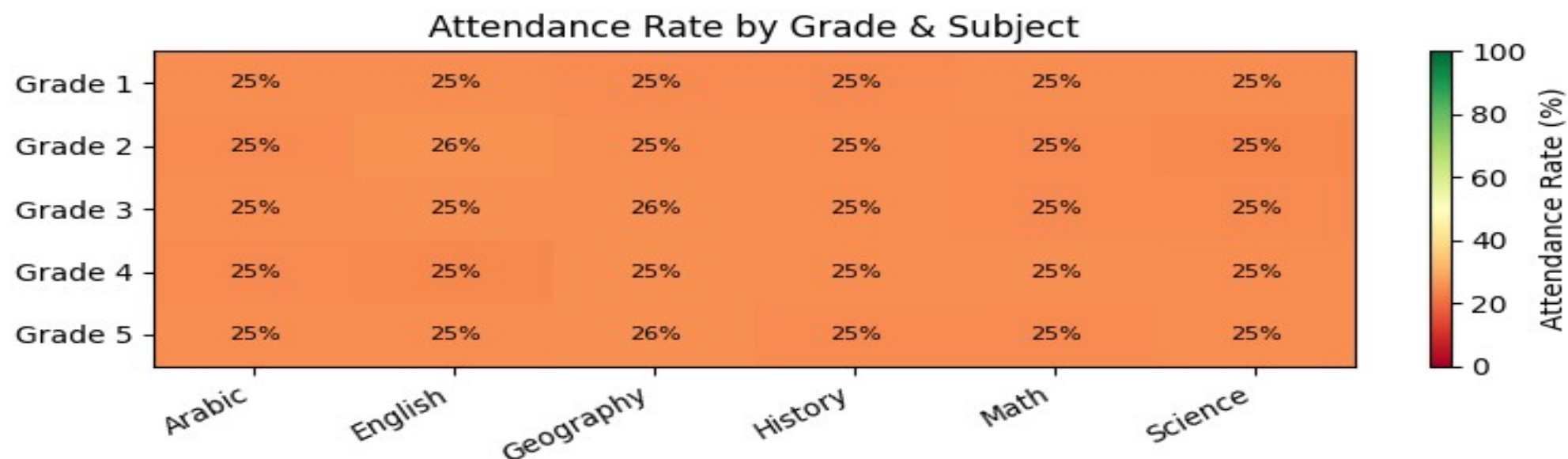
Average exam scores are consistent across all six subjects, sitting in the mid-70s with no subject standing out as substantially harder or easier.

~75

average score
across all subjects



Results: Attendance Patterns



Attendance rate by grade level and subject shows no systematic pattern — no grade or subject sees meaningfully higher or lower attendance than any other.

Key Insights



≈

Scores are consistent

Mid-70s average across every subject, no outliers.



0

No correlation found

Attendance and homework completion show ~0 correlation with exam score, in every subject.



—

Flat attendance trend

Monthly attendance rate stays roughly flat over the full period covered.



◆

Likely synthetic data

The absence of any attendance/homework → performance relationship is consistent with randomly generated data, not real classroom outcomes.

Conclusion & Next Steps

This project delivered a clean, merged, analysis-ready dataset and a full set of subject and attendance visualizations — the descriptive foundation for deeper analysis.

Suggested next steps

- Predictive modeling (regression on exam score, or classification of at-risk students) — out of scope for this submission
- Validate findings against real (non-synthetic) classroom data
- Extend the attendance/homework analysis with time-of-year or subject-difficulty controls